

Problem 5.3

Show that Banker's Rule is always more favorable to the lender than exact simple interest.

Let P be the principal, r the annual interest rate, and d the actual number of days.

Using exact simple interest, the interest is

$$I_{\text{exact}} = Pr \left(\frac{d}{365} \right).$$

Using Banker's Rule, the interest is

$$I_{\text{banker}} = Pr \left(\frac{d}{360} \right).$$

Since d , P , and r are positive, it is enough to compare

$$\frac{d}{365} \quad \text{and} \quad \frac{d}{360}.$$

Because $360 < 365$, dividing the same positive number d by the smaller denominator gives the larger value. Hence,

$$\frac{d}{365} < \frac{d}{360}.$$

Multiplying both sides by the positive quantity Pr , we obtain

$$Pr \left(\frac{d}{365} \right) < Pr \left(\frac{d}{360} \right).$$

Therefore,

$$I_{\text{exact}} < I_{\text{banker}}.$$

So, Banker's Rule always produces more interest than exact simple interest, which means it is always more favorable to the lender.